A Comparative Study of Wav2Vec 2.0 and Compact CNN Models for Cross-Corpus Speech Dysfluency Detection

Abstract. We compare two model families for cross-corpus speech dysfluency detection: self-supervised Wav2Vec 2.0 and compact mel-spectrogram CNNs. F1, the standard metric here, depends on the decision threshold as well as the model, and thresholds calibrated on one corpus need not transfer. Seven configurations are compared: five independent binary Wav2Vec2 classifiers (arm01); two shared-backbone multitask variants frozen for 3 and 20 epochs (arm02, arm03); and four compact mel-spectrogram CNNs with different sequence aggregators (arm04–arm07). All are trained on 29,296 clips from SEP-28K and UCLASS, then evaluated under a clip-level in-distribution split that shares speakers and a speaker-independent cross-corpus split on 53 English Project Boli recordings. In-distribution, arm01 attains the highest measured macro F1 (0.570, 95% CI 0.554–0.587, validation-tuned thresholds), with arm02 close behind (0.556, 0.540–0.572) at $\approx$4.9$\times$ fewer parameters; the intervals overlap. Cross-corpus, the ordering inverts: an all-positive classifier attains macro F1 0.550 (0.490–0.604), which nothing exceeds, because the CNNs reach their F1 by labelling nearly every external clip positive (macro predicted-positive rate 0.879, 95% CI 0.849–0.909, against a base rate of 0.408) while the Wav2Vec2 configurations collapse toward all-negative (0.079–0.102). Threshold-free metrics reverse the ranking again: the task-adapted Wav2Vec2 configurations rank external clips best (macro AUROC 0.620–0.621, AUPRC 0.546–0.557) and the CNNs lie near chance. The predicted-positive rate shows the CNN operating points already sat far above the in-distribution base rate, so the CNN advantage under thresholded F1 is consistent with operating-point drift and does not imply superior ranking. Thresholded F1 alone therefore cannot support cross-corpus claims for this task. Each configuration was trained once; intervals reflect clip-level resampling, and the external analysis is exploratory.

Keywords: Speech dysfluency detection · Wav2Vec 2.0 · spectrogram CNNs · cross-corpus evaluation · distribution shift · speaker-independent evaluation.

1 Introduction

Stuttering affects approximately 1% of the global adult population and up to 5% of children [16]. It is characterized by involuntary repetitions of sounds and words, abnormal prolongations, and pauses in speech flow (blocks), and it affects the psychosocial well-being of those who stutter. Clinical assessment by

speech-language pathologists relies on subjective listening and waveform inspection, which is time-consuming, subject to inter-rater variability, and does not scale; this has motivated substantial work on automatic stuttering identification [14]. Self-supervised speech representation learning, and Wav2Vec 2.0 in particular [2], has since produced strong downstream results from raw audio. Applied to dysfluency detection, however, this line of work is dominated by a single evaluation style: models are trained and tested on a common corpus, and performance is summarized with per-class and macro F1 at a threshold tuned on held-out data from that same corpus. F1 at such a threshold measures two things at once: the quality of the model's ranking of clips, and the suitability of a threshold calibrated on data resembling the test data. Only the first should transfer to new speakers and corpora. The two can move in opposite directions under shift: a threshold that maximizes validation F1 in-distribution may fall in a degenerate region of the ROC curve once the score distribution changes, leaving F1 dominated by class base rates rather than discrimination. The external corpora available for this task are small enough that a single operating point can dominate the reported result.

Cross-corpus and speaker-independent evaluation is not new in principle. Bayerl et al. [5] studied cross-corpus stuttering detection as a multi-label problem, and Bayerl et al. [6] showed that the choice of dataset partitioning materially changes reported results. The interaction between the two metric families under shift is less well characterized: how much of a cross-corpus F1 difference is attributable to ranking quality, and how much to operating-point drift. Without that separation, comparisons between representation families risk being read as statements about generalization when they concern threshold placement. We address this with a controlled comparison of seven configurations, Wav2Vec 2.0 models and compact spectrogram CNNs, under matched protocols, reporting thresholded and threshold-free views side by side. Our research questions are as follows.

- **RQ1**: Under conventional clip-level in-distribution evaluation, how do the Wav2Vec2-based and compact spectrogram CNNs compare in macro and per-class F1?
- **RQ2**: On a speaker-independent external corpus, how do their measured F1 scores and their ordering change?
- **RQ3**: How much of the cross-corpus F1 behaviour is explained by operating-point drift (a mismatch between predicted-positive rate and class base rate) rather than by a change in ranking quality?
- **RQ4**: Do threshold-free metrics (AUROC, AUPRC) give a materially different account of cross-corpus behaviour than thresholded F1?

No configuration generalizes across corpora reliably (Section 5). Our contributions are: (i) a matched comparison of Wav2Vec 2.0 and compact spectrogram CNN configurations under in-distribution and speaker-independent cross-corpus protocols; (ii) an explicit separation of ranking quality from operating-point placement, using the predicted-positive rate measured on both splits under identical thresholds, which shows the two metric families yielding opposite orderings

on the external set; (iii) the observation that no configuration exceeds an all-positive baseline on external F1, a negative result; and (iv) a per-class analysis identifying interjection as easiest and block as hardest.

2 Related Work

2.1 Detection Approaches

Early systems used handcrafted features (jitter, shimmer, spectral flux, formant trajectories) with SVMs, HMMs, or GMMs [14] and generalized poorly across speakers, accents, and conditions. End-to-end deep learning removed the manual feature-engineering step: CNNs on mel-spectrograms with recurrent layers showed early promise for multi-type dysfluency detection [12], as did RNNs and time-delay networks for temporal dependencies [13]. These remain attractive where labelled data and compute are scarce, and define the compact end of the capacity range compared against here. Self-supervised pre-training has since become dominant: Wav2Vec 2.0 contrasts quantized latent speech units against masked waveform spans and generalizes across languages when pre-trained multilingually (XLSR-53 [9]), HuBERT [11] predicts masked hidden units from offline clustering, XLS-R [1] scales the objective to 128 languages, and WavLM [8] adds denoising objectives. Bayerl et al. [4] applied Wav2Vec2 to multi-task stuttering detection on FluencyBank, reporting macro F1 of 0.56–0.63 across five types, and to UCLASS therapy sessions [7]; on the standardized SEP-28k-E partition per-class F1 varies widely by type (interjection 0.78, prolongation 0.53, block 0.30) [15]. Common to these models is that their embeddings encode linguistic content together with paralinguistic attributes such as speaker identity and recording condition; how much corpus-specific information a representation carries is the trade-off our comparison probes.

2.2 Corpora and Metrics Under Shift

The three corpora used here (Section 3) differ in speakers, recording conditions, label definitions, and annotation granularity, so generalization to unseen conditions is difficult to assess. Language is not among the differences that our evaluation actually exercises, because the ingested Boli portion is English-only (Section 3). Bayerl et al. [5] framed cross-corpus detection as a multi-label problem, motivated by dysfluent events co-occurring rather than appearing in isolation, and showed that the partitioning scheme changes reported results substantially [6]; most of this literature still reports F1 at a threshold chosen on in-distribution data, the practice whose adequacy we examine. Our study is methodological: we hold the model set fixed, vary the evaluation protocol, and report thresholded and threshold-free metrics for the same predictions.

F1 is the harmonic mean of precision and recall at a chosen operating point, so its value depends on the threshold as well as on the score ranking. AUROC summarizes ranking over all thresholds but can look optimistic when positives

are rare, because a large negative class inflates the area; AUPRC weights precision by the positive class, is the more informative of the two under imbalance, and has the positive-class prevalence rather than 0.5 as its chance level. Predictive calibration is a separate property that we do not measure [10]. We report, for each configuration and protocol, thresholded F1, threshold-free AUROC and AUPRC, the AUPRC lift over the prevalence prior, and the predicted-positive rate (PPR): at a fixed threshold, a PPR far from the base rate indicates displaced operating-point placement rather than lost discriminative ability. Because the same thresholds are applied to both evaluation sets, the difference in PPR between them measures drift directly rather than by inference.

3 Corpora and Models

The system is a clip-level classification pipeline that predicts which of the five dysfluency types are present in a 3-s audio window. All models share one preprocessing module and are exposed through a common registry API.

3.1 Source Corpora and Clip Construction

SEP-28K contributes 32,321 clips (28,177 podcast clips and a 4,144-clip reannotation of adult FluencyBank speech); UCLASS contributes 4,712 therapy-session clips in SEP-28K format; and Project Boli contributes 54 clips with word-level annotations and timestamps; Boli is confined to the test split, so no external speaker is seen in training. The three are merged into 37,087 manifest rows, with per-clip interval CSVs recording events as `(start_sec, end_sec, type)` tuples; clips whose audio is missing or empty are dropped, leaving 36,674 usable clips. The 36,620 non-Boli clips are partitioned 80/10/10 at random under seed 42 into 29,296 training, 3,662 validation, and 3,662 test clips, and all 54 Project Boli clips are then pinned to the test split, giving a test partition of 3,716 rows. Audio is resampled to 16 kHz mono, DC-removed, peak-normalized, and silence-trimmed.

Evaluation does not iterate over manifest rows. It enumerates the audio files present in a split directory and keeps those that have a matching label CSV and a non-empty body, so the number of scored clips is the number of distinct scored recordings rather than the number of rows admitted. On the test partition this yields 3,715 scored clips, of which 53 are Project Boli; the difference between the 54 admitted rows and the 53 scored recordings is one row that this filter does not admit. The same rule applied to the external evaluation set is what produces the Boli count of 53 used throughout this paper.

Construction of the Project Boli subset The external subset is derived by an explicit four-step selection procedure from the released Boli corpus. First, all `.txt` files in the `Transcripts` directory are enumerated in sorted order; each is a word-level annotation list whose lines carry a start time, an end time, and a dysfluency code. Second, each transcript is paired with an audio file by name: a

direct filename lookup is attempted, then a mapped lookup through a fixed task table, then a recursive basename search; a transcript with no resolvable audio is discarded. Third, the paired transcript is parsed line by line, lines with fewer than three fields are skipped, and the remaining codes (*B*, *PR*, *SR*, *WR*, *IN*) are mapped to block, prolongation, sound repetition, word repetition, and interjection; a transcript yielding no recognized code is discarded. The 54 surviving transcripts each contribute one manifest row whose five binary labels are set by whether any annotated word carries the corresponding code. Fourth, all 54 rows are pinned to the test split and never seen in training; scoring then enumerates distinct audio filenames in that split, so the 54 rows resolve to 53 distinct scored recordings, and all external results are computed on those 53 clips.

Because the pairing key is the audio filename, the scored unit is the distinct recording rather than the annotation row, which is the mechanism behind the 54-to-53 reduction; the reports retain only the resulting counts, so we cannot say which filter removed the single row. Three properties of the released corpus matter for interpreting the external results. First, each annotated file is built around a single stuttered word [3], so the external set is enriched for dysfluent content by construction and is not a sample of naturalistic speech. Second, the released corpus is multilingual: the Boli paper reports 28 participants who stutter (22 Hindi, 2 Telugu, 2 Bengali, 1 Marathi, and 1 Assamese mother tongue), contributing roughly 2.5 h of audio. Our ingestion path resolves only the English task codes of that release, so all 54 clips we ingest are English and the cross-corpus protocol below changes speakers, recording pipeline, and annotation convention, but not language: it is a cross-corpus evaluation and not a cross-lingual one, and cross-lingual robustness remains untested by these results. Third, the retained audio segment is the first 3 s of each file (Section 3.3), whereas the Boli label derives from the whole file, so a recording longer than 3 s carries a label that need not be audible in the segment the model receives.

3.2 Annotation Harmonization

The three corpora use different conventions, which we map onto five common binary clip-level labels: prolongation, block, sound repetition, word repetition, and interjection. SEP-28K labels are adopted as-is, with a clip positive for a type when at least two annotators marked it (`SEP28K_MIN_VOTES` = 2); requiring two votes is a substantive choice, since accepting any single vote inflates positives several-fold. Because each UCLASS clip is a 3-s window centered on a stutter event, each present label is materialized as a weak interval of half-width 0.25 s centered at 1.5 s, i.e. spanning 1.25–1.75 s. Project Boli provides word-level tokens labeled *B*, *PR*, *SR*, *WR*, and *IN*, mapped through a fixed code table to the same five classes as described in Section 3.1, a clip being positive for a class when any annotated word carries it. All models therefore consume identical five-dimensional binary targets.

3.3 Preprocessing and the 3-Second Window

SEP-28K clips are natively 3 s, so a 3-s maximum window matches the predominant corpus; because the task is clip-level, an event whose acoustic extent crosses the window boundary does not alter the target. Audio longer than 3 s is truncated to its first 3 s and shorter audio is zero-padded, so every configuration receives an identically shaped input. Two consequences recur below: silence trimming risks suppressing the pause that defines a block, and for any Boli recording longer than 3 s the label is derived from the whole file while the retained segment is its first 3 s, so a positive label need not be present in the audio the model receives (Section 6.2).

3.4 Model Configurations

Seven configurations are evaluated. Trainable parameter counts are 472,845,450 for arm01 (five independent Wav2Vec2 binary classifiers), 97,332,362 each for arm02 and arm03 (Wav2Vec2 multitask, backbone frozen for 3 and 20 epochs), and 341,322, 274,242, 456,906, and 539,594 for arm04–arm07 (CNN with pooling, single-head CNN with pooling, CNN with LSTM, and CNN with transformer). The CNNs therefore hold 0.3–0.6% of the parameters of arm01. The evaluation artifacts report slightly larger figures for the CNNs—342,030, 274,950, 457,614, and 540,302—because they count every element of the stored checkpoint, including 708 BatchNorm buffer elements (running statistics) that are not trainable; we quote the trainable counts here. Inference latency, throughput, and peak GPU memory were not measured and are not inferred from these counts (Section 5.5).

Model descriptions. arm01 handles each dysfluency type with a separate binary classifier on `facebook/wav2vec2-base`, 94,569,090 parameters each: raw audio → encoder → a learned projection of the frame-level hidden states (`Linear`(768→256)) → mean pooling over time → a linear classifier (`Linear`(256→2)) → softmax. Two details are easy to assume otherwise and are therefore worth stating: the projection precedes the pooling rather than following it, and the head contains no dropout layer. No attention mask is passed by the pipeline, so the temporal reduction is an unmasked mean over frames; because every clip is zero-padded or truncated to a fixed 3 s, all clips present the same number of frames. Each classifier is trained independently with focal loss ($\gamma = 2.0$), AdamW (lr $= 3 \times 10^{-5}$, weight decay $= 0.01$), a 500-step linear warmup, and mixed precision, the backbone frozen for the first 3 epochs; once the backbone is unfrozen it is trained with a reduced learning rate of 3×10^{-6}, one tenth of the head rate. arm02 and arm03 instead feed five parallel per-class heads from one shared encoder, each an MLP on a temporal mean pool (`Linear`(768→768) → Tanh → `Linear`(768→2)), minimizing the summed focal loss, for 97.3M parameters in total, only ≈2.8M more than one independent classifier, because the backbone is shared. The two differ only in that arm03 freezes the backbone for 20 epochs instead of 3, isolating freezing duration;

the 3-epoch freezing schedule and the post-unfreeze learning-rate reduction of arm01 apply to both.

arm04–arm07 share a spectrogram backbone and differ in the sequence aggregator. Input is a log-mel spectrogram with 128 Mel bands, a 2048-point FFT, and hop length 512. The encoder is four convolutional blocks with 32, 64, 128, and 128 channels, each with a 3×3 kernel (padding 1), batch normalization, ReLU, a $(2, 1)$ max-pool in all but the last block, and 2-D dropout at rate 0.4. The three pools halve the frequency axis three times and leave the temporal axis at 94 frames, so the encoder output is a $128 \times 16 \times 94$ tensor (channels $\times$ mel bins $\times$ frames). This is aggregated by global mean pooling plus a `Linear`(128→128) projection (arm04); a single-layer unidirectional LSTM with hidden size 128 over the frequency-pooled time axis, using the final hidden state (arm06); or a one-layer transformer encoder with model dimension 128, four heads, feed-forward width 512, and dropout 0.4, followed by temporal mean pooling (arm07). Five per-class two-layer MLP heads (`Linear`(128→128) → Tanh → `Linear`(128→2)) produce the logits. The CNN arms are trained from scratch, so no backbone freezing or discriminative pre-training applies, and their objective differs from that of the Wav2Vec2 arms: instead of focal loss they use binary cross-entropy on the positive logit with a per-class positive weight set to the negative-to-positive class ratio of the training pool, with the per-head losses summed over the configured classes. That ratio is computed over the whole training directory rather than over the training subset alone, a difference from the multitask Wav2Vec2 trainer that we record but do not expect to matter. The two families therefore differ in training objective as well as in representation, which we note as a confound when comparing them. arm05 is the single-head variant of the pool model: the head block is built once per configured class rather than five times, so each arm05 model carries the same two-layer head but only a single instance of it, and one such model is trained per dysfluency type (five models in total), mirroring the five-binary Wav2Vec2 design at a much smaller scale.

4 Experimental Setup

4.1 Training Protocol

All models are trained on a single NVIDIA GPU with seed = 42, 20 epochs and early stopping (patience = 5), mixed precision, batch size 8 (binary classifiers) or 16 (multitask and CNN), and 3 s audio at 16 kHz. Training reads the training directory only: it derives its own held-out portion from it, a 20% stratified subsample (5,857 clips, seed 42) that serves as the early-stopping and threshold-tuning set, so the parameters are fitted on 23,439 clips and the prepared validation split is never opened. No gradient clipping is applied; an earlier configuration that clipped the total gradient norm was abandoned because, on a 94.6M-parameter model with a randomly initialized head, the clip dominated the head's early gradient signal. The checkpoints evaluated here were trained without input augmentation, so we claim no augmentation effect. Each configuration is trained once, so every reported value is a single-run point estimate.

4.2 Threshold Selection

Thresholds are tuned on a held-out 20% stratified subsample of the training pool (5,857 clips, seed 42) by sweeping 17 fixed values from 0.1 to 0.9 in steps of 0.05 and, per class, selecting the value maximizing F1 on that subsample rather than Youden's J. That subsample is also the early-stopping set, so the tuned thresholds are mildly optimistic; we treat this as a limitation rather than as test-set leakage, since the test split is never used for any selection. The prepared 3,662-clip validation split is not consumed by the training or evaluation code and is unused in every number reported here. arm02 uses thresholds of 0.55, 0.35, 0.40, 0.35, and 0.45 for prolongation, block, sound repetition, word repetition, and interjection. No test-set or cross-corpus threshold fitting is performed anywhere in this paper; the same thresholds are applied unchanged to the external set, the transfer that Sections 5 and 6 examine.

4.3 Reported Metrics

For each configuration, protocol, and class we report precision, recall, F1, AUROC, and AUPRC, plus macro averages (the unweighted mean over the five classes), which keep classes of differing prevalence from being dominated by the most frequent; the behaviour of these metrics under shift is discussed in Section 2. AUPRC is reported as a lift over its chance level, the positive-class prevalence. Precision and recall are omitted from the main tables, having fixed the threshold in advance, but are recoverable from F1 and the PPR with the confusion counts.

4.4 Evaluation Protocols

In-distribution, clip-level (Test). The test partition holds 3,716 manifest rows, of which 3,662 come from SEP-28K and UCLASS and 54 from Project Boli. The partition operates on clips, not speakers, so speakers may appear in both training and test for the SEP-28K and UCLASS portions; these numbers are an upper-bound benchmark under near-distribution conditions and are not speaker-independent. Results are reported over the 3,715 scored clips, of which the 53 scored Boli recordings (≈1.4%) are out-of-domain. The macro base rate on this split is 0.143.

Speaker-independent, cross-corpus (Boli). The 53 scored Boli recordings are unseen during training, with different speakers, recording conditions, and label-construction procedure (Section 3). Per-class support is small: PR 16, BL 35, SR 34, WR 14, IN 9, giving a macro base rate of 0.408. We treat all external conclusions as exploratory.

Two trivial baselines accompany the models: the *all-positive* classifier, which labels every clip positive for every class, and the *random-at-rate* classifier, which labels clips positive with probability equal to the class base rate. All-positive is the more informative under heavy imbalance, because it attains recall 1 by

construction and its F1 is determined entirely by the base rate, so a model that does not exceed it has not demonstrated usable discrimination; the expected macro F1 of random-at-rate equals the macro base rate.

4.5 Statistical Uncertainty

Uncertainty is quantified by nonparametric bootstrap resampling of clips: for each reported quantity we resample the evaluated clips with replacement 10,000 times, recompute the metric, and take the 2.5th and 97.5th percentiles. Resampling is unstratified, so it propagates variability in class prevalence as well as in the decisions; on the 53-clip external set the former dominates for the trivial baselines. F1 at a fixed threshold and the predicted-positive rate are functions of the confusion counts alone, so their bootstrap operates exactly on the stored per-class confusion matrices. These intervals capture clip-level sampling variability only: each configuration was trained once, so they exclude training seed, and overlapping intervals should not be read as a real difference in model quality. AUROC and AUPRC are reported as point estimates without intervals, because both depend on the full per-clip score vector, which was not retained after evaluation, so no bootstrap of them can be formed; we therefore treat the threshold-free ordering as directional evidence and the external results as exploratory.

5 Results

5.1 In-Distribution Results (RQ1)

Table 1 reports macro F1 for all seven configurations under fixed (0.5) and tuned thresholds on both evaluation sets, with the two trivial baselines. Parenthesized intervals are 95% bootstrap confidence intervals (Section 4.5).

Table 1. Macro F1 (unweighted mean over the five classes) for all seven configurations and two trivial baselines, with 95% bootstrap confidence intervals on the tuned columns. "Test" is the clip-level in-distribution split (3,715 scored clips, speakers shared with training); "Boli" is the speaker-independent cross-corpus split (53 scored recordings). "0.5" applies a fixed threshold; "tuned" applies the per-class validation-F1-optimal thresholds of Section 4, unchanged on Boli. The two baselines are untuned, so their columns coincide. Intervals reflect clip-level resampling only and exclude seed variation.

Arm	Test F1 (0.5)	Test F1 (tuned)		Boli F1 (0.5)	Boli F1 (tuned)	
arm01_5x_w2v2	0.573	0.570	(0.554–0.587)	0.254	0.292	(0.169–0.391)
arm02_mt_frz3	0.542	0.556	(0.539–0.572)	0.162	0.238	(0.127–0.332)
arm03_mt_frz20	0.141	0.378	(0.362–0.394)	0.046	0.214	(0.119–0.298)
arm04_cnn_pool	0.149	0.256	(0.245–0.266)	0.351	0.332	(0.254–0.398)
arm05_cnn_single	0.212	0.262	(0.251–0.272)	0.450	0.476	(0.389–0.547)
arm06_cnn_lstm	0.242	0.249	(0.240–0.257)	0.451	0.508	(0.440–0.566)
arm07_cnn_tf	0.269	0.271	(0.259–0.283)	0.421	0.345	(0.260–0.419)
All-positive	0.247	0.247	(0.239–0.254)	0.550	0.550	(0.490–0.604)
Random@rate	0.143	0.143		0.408	0.408	

Under the clip-level protocol, arm01 attains the highest measured test macro F1 at tuned thresholds and arm02 follows closely: 0.570 (0.554–0.587) against 0.556 (0.539–0.572), with substantially overlapping intervals on a split of 3,715 clips. Since the intervals describe clip sampling alone and each configuration was trained once, we report the pair as indistinguishable and do not treat arm01 as the better model. Both exceed the remaining configurations by a wide margin, their intervals lying far above those of every CNN. The CNNs reach macro F1 of 0.25–0.27 at tuned thresholds, above the random-at-rate baseline (0.143) but only marginally above the all-positive baseline (0.247), and in the case of arm06 the two intervals overlap. Freezing duration is the one architectural factor with a large and unambiguous effect: arm03, which freezes the backbone for 20 epochs rather than 3, falls to 0.378 (0.362–0.394) tuned macro F1, a relative reduction of ≈32% from arm02, with non-overlapping intervals, which is evidence that task adaptation of the encoder is necessary here. Its fixed-0.5 score of 0.141 also falls below the random-at-rate baseline, a further sign that its scores are poorly distributed rather than merely poorly ranked.

5.2 Cross-Corpus Results at Tuned Thresholds (RQ2)

On the 53-clip external set, the all-positive baseline attains macro F1 0.550 (0.490–0.604), which no configuration exceeds; the best model, arm06, reaches 0.508 (0.440–0.566). The two intervals overlap, so this is a comparison of point estimates rather than a separated difference, and the ordering among the CNNs is likewise unresolved: the four CNN intervals span 0.254 to 0.566 and overlap extensively. The one clear separation is between the best CNN and the two task-adapted Wav2Vec2 configurations, whose intervals (0.169–0.391 and 0.127–0.332) lie below that of arm06. Table 2 gives the per-class breakdown.

Table 2. Per-class F1 on the speaker-independent cross-corpus set (53 scored recordings) at the validation-tuned thresholds, transferred unchanged. The all-positive row labels every clip positive, giving per-class F1 $2s/(n+s)$ for support s and $n = 53$; no configuration exceeds it in macro F1. The macro column carries a 95% bootstrap interval; per-class intervals are omitted because with support between 9 and 35 they are too wide to inform the comparison (e.g. arm06 block, 0.795, 95% CI 0.691–0.873).

Arm	PR	BL	SR	WR	IN		Macro
arm01_5x_w2v2	0.400	0.250	0.256	0.125	0.429	0.292	(0.169–0.391)
arm02_mt_frz3	0.200	0.154	0.256	0.182	0.400	0.238	(0.127–0.332)
arm03_mt_frz20	0.200	0.233	0.182	0.182	0.273	0.214	(0.119–0.298)
arm04_cnn_pool	0.308	0.491	0.182	0.387	0.290	0.332	(0.254–0.398)
arm05_cnn_single	0.326	0.765	0.508	0.431	0.348	0.476	(0.389–0.547)
arm06_cnn_lstm	0.492	0.795	0.542	0.418	0.290	0.508	(0.440–0.566)
arm07_cnn_tf	0.316	0.320	0.490	0.267	0.333	0.345	(0.260–0.419)
All-positive	0.464	0.795	0.782	0.418	0.290	0.550	(0.490–0.604)

The CNN rows appear to show that compact spectrogram models "generalize best". That reading is incorrect, and Table 3 shows why. The CNNs reach their higher F1 on block, word repetition, and interjection by labelling nearly every

external clip positive: arm06 applies the positive label to all 53 clips for each of those three classes, so its per-class F1 approaches, and for interjection exactly equals, the value a predictor with no learned discrimination attains at the same support. Its macro predicted-positive rate is 0.879, more than twice the 0.408 base rate, while the Wav2Vec2 configurations move the opposite way at PPR 0.079–0.102, leaving their F1 limited by near-zero recall. Both behaviours follow from where a fixed threshold lands after the score distribution shifts, and neither is evidence about ranking quality. The finding is negative: at thresholds never fitted to the external set, no configuration demonstrates usable dysfluency discrimination, and the apparent advantage of the CNN family under thresholded F1 is consistent with substantial operating-point drift and does not by itself indicate superior ranking quality.

5.3 Threshold-Free Cross-Corpus Results and Operating-Point Drift (RQ3, RQ4)

Because F1 conflates ranking with threshold placement, we also report threshold-free quantities for the same predictions (Table 3), together with the predicted-positive rate measured on the in-distribution split under the same thresholds. AUPRC lift is AUPRC divided by the macro base rate of 0.408, the AUPRC of a random predictor.

Table 3. Threshold-free macro metrics and predicted-positive rate (PPR) from the same predictions as Table 2. AUROC and AUPRC are point estimates; AUPRC lift is relative to the macro base rate 0.408, its chance level. PPR is the mean fraction of clips labelled positive at the tuned thresholds, so it measures where the operating point landed rather than performance, and it carries a 95% bootstrap interval. The two PPR columns apply identical thresholds to the two evaluation sets, so their difference measures drift directly. Base-rate rows give the in-distribution (0.143) and external (0.408) priors.

Arm	AUROC	AUPRC	AUPRC lift	PPR (Test)		PPR (Boli)
arm01_5x_w2v2	0.620	0.546	1.34	0.134	0.079	(0.049–0.113)
arm02_mt_frz3	0.621	0.557	1.37	0.159	0.102	(0.068–0.140)
arm03_mt_frz20	0.538	0.443	1.09	0.195	0.204	(0.158–0.253)
arm04_cnn_pool	0.473	0.412	1.01	0.572	0.574	(0.528–0.619)
arm05_cnn_single	0.497	0.445	1.09	0.583	0.615	(0.566–0.664)
arm06_cnn_lstm	0.522	0.419	1.03	0.845	0.879	(0.849–0.909)
arm07_cnn_tf	0.500	0.412	1.01	0.365	0.438	(0.381–0.494)
Base rate (random)	0.500	0.408	1.00	0.143		0.408

The threshold-free view inverts the thresholded ordering once more. arm01 and arm02 rank the external clips best, with macro AUROC 0.620–0.621 and macro AUPRC 0.546–0.557, a lift of ≈1.3–1.4× over the prevalence prior; arm03, the over-frozen variant, sits near chance; and the CNNs are at or barely above chance, with macro AUROC 0.473–0.522 and macro AUPRC 0.412–0.445. The advantage is modest: these are point estimates without intervals, and on 53 clips with at most 35 positives per class and 9 in the worst case, a macro AUROC

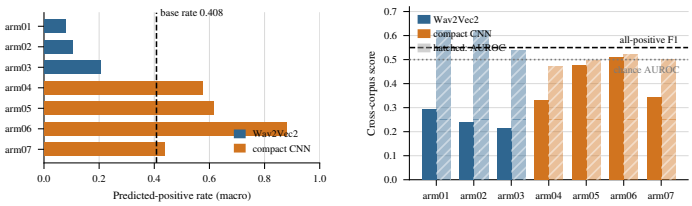
of 0.62 against 0.52 is well within the range that resampling could plausibly reduce to nothing. We therefore state the result directionally: the threshold-free metrics place the task-adapted Wav2Vec2 configurations above the CNNs, and no configuration comes near reliable cross-corpus detection.

The same table quantifies the mechanism behind that disagreement, which is our answer to RQ3. Because the two PPR columns apply identical thresholds to the two evaluation sets, the comparison is a measurement rather than an inference. Three facts follow.

First, the operating points are far from the prevailing base rate in both protocols: for six of the seven configurations the 95% interval for external macro PPR excludes the external base rate of 0.408, the sole exception being arm07 (0.381–0.494). The three Wav2Vec2 configurations sit far below the external base rate, at 0.079–0.204, and the four CNNs above it, at 0.438–0.879.

Second, the CNN operating points were already displaced before the shift. On the in-distribution split, where the base rate is 0.143, the CNNs label 0.365–0.845 of clips positive, a factor of 2.6–5.9 above prevalence, while reaching only 0.249–0.271 macro F1. The external F1 they appear to win is thus not produced by the shift: the same over-positive operating point that cripples their in-distribution F1 becomes advantageous externally only because the Boli base rate (0.408) is nearly three times the in-distribution one. In relative terms their operating point is *less* displaced externally than in-distribution (for arm06, 5.9 times the base rate on the test split against 2.2 times on Boli), the opposite of the drift a generalization failure would produce.

Third, the task-adapted Wav2Vec2 configurations move the other way under the same thresholds, from 0.134 and 0.159 in-distribution to 0.079 and 0.102 externally, toward all-negative. Their low external F1 is the mirror image of the CNNs' behaviour; neither is a statement about ranking quality, because AUROC and AUPRC do not move with the threshold and so order the two families the other way round.



(a) predicted-positive rate (b) macro F1 vs. AUROC

Fig. 1. The two metric families disagree, and the mechanism is visible. (a) Operating-point drift on the cross-corpus set (RQ3): macro predicted-positive rate at the in-distribution-tuned thresholds, against the 0.408 macro base rate (dashed), as in Table 3. (b) RQ2 against RQ4: cross-corpus macro F1 (solid) against macro AUROC (hatched); no configuration reaches the all-positive F1 line, and the two orderings are near-reversals.

5.4 Per-Class Analysis

Table 4 gives per-class metrics for arm02 on the clip-level test split at its tuned thresholds.

Table 4. Per-class metrics for the multitask Wav2Vec2 model (arm02) on the in-distribution test split (3,715 scored clips) at the validation-tuned thresholds 0.55, 0.35, 0.40, 0.35, and 0.45 for prolongation, block, sound repetition, word repetition, and interjection. Base rate is support divided by 3,715, the chance level of AUPRC for that class. F1 carries a 95% bootstrap interval.

Class	F1 (tuned)	AUROC	AUPRC	Support	Base rate
Prolongation	0.457 (0.407–0.503)	0.845	0.491	400	0.108
Block	0.429 (0.397–0.460)	0.768	0.444	553	0.149
Sound Rep.	0.621 (0.584–0.656)	0.878	0.624	494	0.133
Word Rep.	0.499 (0.456–0.541)	0.855	0.550	366	0.099
Interjection	0.773 (0.750–0.795)	0.935	0.848	834	0.225
Macro	**0.556 (0.540–0.572)**	**0.856**	**0.592**	—	**0.143**

Interjection is the easiest class, at F1 0.773 (0.750–0.795) with AUROC 0.935, and block the hardest, at F1 0.429 (0.397–0.460) with AUROC 0.768; the two intervals are well separated. Read against their own chance levels, the AUPRC values in Table 4 are 2.99–5.59 times the base rate, so the ranking signal is real in-distribution even where thresholded F1 is low. One reading is that interjections are lexically stereotyped fillers whose acoustic form differs from surrounding fluent speech, whereas blocks are pauses close to ordinary hesitation that depend on prosodic and contextual information a 3-s window and a mean-pooled representation may not preserve; the two mechanisms are not separable here, and the block result fits the window and trimming artifacts of Section 3 equally well.

For every class the AUROC exceeds the tuned F1, most markedly for prolongation (AUROC 0.845 against F1 0.457): the models order dysfluent clips better than the tuned thresholds convert that ordering into decisions. Whether this reflects miscalibration or a mismatch between F1-optimal and accuracy-optimal operating points cannot be settled here, because we do not measure calibration. The arm01 block classifier produces the most false negatives (343) and the most false positives relative to its support (284), the same pattern visible in its per-class F1 and AUROC.

5.5 Computational Cost and Relation to Published Results

Parameter count, storage size, computational cost, and runtime cost are distinct: the first two follow from the weights and a stated precision, the last must be measured. We therefore report parameter counts and their fp32 storage in Section 3 but not latency, throughput, or peak GPU memory, and we do not infer them from parameter counts, which give fp32 storage of 1.89 GB for arm01 against 1.1–2.2 MB for the CNNs. Within the Wav2Vec2 family we would select the multitask configuration for operational reasons: 97.3M parameters against 472.8M because the encoder is shared rather than replicated five times, and

one forward pass per clip instead of five. It does not measurably beat arm01. Our best in-distribution macro F1, 0.570 (five-binary) and 0.556 (multitask), falls within the 0.56–0.63 reported for Wav2Vec2 on FluencyBank [4], and our per-class ordering matches the 0.30 (block) to 0.78 (interjection) span on SEP-28k-E [15]; this is an ordering comparison, not a numerical one, since our split shares speakers between training and test and the corpora, label definitions, and threshold strategies differ [6].

6 Discussion

6.1 What the In-Distribution Results Show

The Wav2Vec2 configurations capture in-distribution signal that the compact CNNs, at roughly 0.3–0.6% of the parameter count, do not. The most likely explanation is representational capacity: a 94M-parameter pretrained encoder supplies features that a four-block CNN trained from scratch on 29K clips cannot match. Capacity, pretraining, and objective are not separable in our design, since the Wav2Vec2 configurations minimize focal loss while the CNNs minimize a class-balanced binary cross-entropy and we vary the architecture family as a whole, so we do not attribute the gap to any single factor. The one factor we can isolate, freezing duration, acts in the opposite direction and is unambiguous: withholding task adaptation from the same encoder costs ≈32% of tuned macro F1, an effect larger than any difference among the CNNs.

6.2 Why the External Results Are Hard to Interpret

Three features complicate every reading of Tables 2 and 3: 53 scored recordings, per-class support as low as 9, and a label-construction procedure unlike that of the training corpora. The bootstrap intervals quantify the first two, and the third is the most consequential. The wide intervals are not a formality: the external macro F1 of the strongest model, 0.508 (0.440–0.566), overlaps that of the all-positive baseline, 0.550 (0.490–0.604), so the negative result of Section 5.2 is a statement about point estimates on a set too small to separate them. The one comparison the data does support cleanly is the operating-point one: the predicted-positive rate intervals in Table 3 exclude the prevailing base rate for six of seven configurations, and the same thresholds applied in-distribution show the CNN operating points were already displaced before the shift. That measurement, not the F1 ranking, is the substantive content of the external evaluation.

Because Boli files are built around an annotated stutter event [3], a positive Boli label reflects a curated event rather than the outcome of sampling continuous speech and applying a majority vote, which is how SEP-28K and UCLASS labels arise. The all-positive baseline consequently scores 0.550 here against 0.247 on the test split, and that gap is at least partly a property of how the external clips were selected rather than of any model. A second complication is the label–window mismatch flagged in Section 3, which depresses recall for reasons unrelated to representation family; we cannot determine how often it occurs, because we did not retain the durations of the evaluated recordings.

6.3 Proposed Explanations

Two explanations the present experiments do not distinguish are stated as proposals rather than findings.

Operating-point drift. The score distributions of all seven configurations are displaced under the shift, so thresholds fixed in-distribution land in different regions of the ROC curve for the two families, and F1 then tracks the predicted-positive rate and the base rate rather than discrimination. The direct evidence is the paired PPR columns of Table 3 and Figure 1(a), in which the families sit on opposite sides of the base rate, together with the ordering flipping when the threshold is removed. Because both PPR columns use the same thresholds, the displacement is measured rather than assumed. What the measurement does not settle is why it occurs, since that would require the score distributions themselves rather than their induced decisions; the disagreement between F1 and AUROC/AUPRC is the expected signature of such displacement, which is exactly when a single thresholded number invites misinterpretation.

Corpus-specificity. The higher-capacity Wav2Vec2 representations may encode corpus-specific characteristics of SEP-28K and UCLASS (speakers, accents, recording conditions, annotation conventions) that transfer weakly to a corpus built by a different procedure, while the coarse spectral patterns retained by a mel-spectrogram CNN transfer more evenly. The threshold-free ordering is consistent with this account, since arm01 and arm02 lead on AUROC and AUPRC while the CNNs sit near chance, but it is equally consistent with the label–window mismatch and label provenance above, and no measurement we have separates these. A probe classifier on frozen embeddings predicting source corpus would be the most direct test; we have not run one.

6.4 Threats to Validity

Statistical power and single-run training. This is the most consequential threat, with two independent sources. Every configuration was trained once from seed 42, so the arm01/arm02 difference of under 0.02 macro F1 is not interpretable even though their bootstrap intervals overlap, and the arm03 degradation carries no training uncertainty either: the intervals in Tables 1–4 describe clip sampling only. Separately, the external set has 53 scored recordings with per-class support from 9 to 35, which is why external F1 comparisons are point estimates and the threshold-free metrics have no intervals. We use "better" and "outperforms" only where a difference exceeds plausible seed and sampling variation.

Uncontrolled properties of the external set, and untested explanations. As detailed in Section 3, the external labels are built differently from the training labels, the ingested clips are English-only and so the external set does not test language shift, and truncation to the first 3 s may decouple labels from retained audio. The window, trimming, and truncation policy is held fixed across all configurations and both protocols, so it does not confound the comparisons

reported here, but it is an uncontrolled absolute factor and a plausible contributor to the block-class difficulty. All tuned F1 values depend on validation-F1-optimal thresholds, and we do not measure calibration, so we cannot distinguish a miscalibrated probability head from a mismatched operating point [10]; the two accounts in Section 6.3 remain proposals. Finally, capacity, pretraining, and objective are not separated, latency and throughput were not measured, and a single external corpus cannot distinguish a general failure of cross-corpus transfer from a property of this corpus.

7 Practical Implications and Conclusion

Model selection for this task cannot rest on in-distribution F1, because that split shares speakers with training and the cross-corpus F1 ordering is dominated by where a fixed threshold lands. Reported results should therefore include, at minimum, a speaker-disjoint or cross-corpus split, threshold-free metrics alongside F1, a trivial all-positive baseline, and the predicted-positive rate at the reported threshold; each would have changed the conclusion we drew from the F1 columns alone.

On RQ1, Wav2Vec2 dominates in-distribution at macro F1 0.55–0.57 against 0.25–0.27 for the CNNs and 0.247 for an all-positive baseline, with the five-binary and multitask variants indistinguishable and excessive freezing costing ≈32% of tuned macro F1. On RQ2 the cross-corpus F1 ordering inverts, with the CNNs appearing to lead, and on RQ3 that inversion is consistent with operating-point drift rather than generalization: at identical thresholds the CNNs label 37–85% and 44–88% of in-distribution and external clips positive against base rates of 0.143 and 0.408, while the Wav2Vec2 configurations fall to 8–10% externally, and no configuration exceeds the all-positive baseline of 0.550. On RQ4 threshold-free metrics reverse the ordering again, placing the task-adapted Wav2Vec2 configurations above the CNNs at macro AUROC 0.620–0.621 against 0.473–0.522. The conclusions we can draw are narrow: F1 alone cannot support cross-corpus claims for this task, and interjection is easier to detect than block under both protocols. We cannot claim cross-corpus deployment, because no configuration exceeded a trivial baseline on the external set of 53 scored recordings with per-class support as low as 9 and at least three confounding properties (Section 3). The cause of the score drift and the modest threshold-free advantage of Wav2Vec2 remain unresolved (Section 6.4).

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